High p-Rank Class Groups of Quadratic Fields

The Case of 5-Rank at Least Two

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Abstract

This project studies the arithmetic statistics of ideal class groups of quadratic number fields. The initial problem is to understand the frequency of imaginary quadratic fields whose ideal class groups have 5-rank at least two. Recent work establishes a quantitative lower bound of order $X^{1/3}/(\log X)^2$. The principal objective of this project is to understand the limitations of the current bound and to seek stronger quantitative results. The project is intended as a long-term investigation of high p-rank phenomena in quadratic class groups.

1 Background

Let K be a number field and let $\mathrm{Cl}(K)$ denote its ideal class group. For a prime p, define the p-rank of $\mathrm{Cl}(K)$ by

$$\mathrm{rk}_p \, \mathrm{Cl}(K) = \dim_{\mathbf{F}_p} \left(\mathrm{Cl}(K)/p \, \mathrm{Cl}(K) \right).$$

The distribution of the p-primary parts of class groups of quadratic fields is a central problem in arithmetic statistics. The Cohen–Lenstra heuristics predict that nontrivial p-torsion, including class groups of large p-rank, should occur with positive frequency for odd primes p. Proving quantitative results of this strength remains far beyond what is currently known in many cases.

We focus on imaginary quadratic fields and define

$$N_{5,2}^{-}(X) = \# \left\{ K/\mathbf{Q} \text{ imaginary quadratic} : |\Delta_K| \leq X, \ \mathrm{rk}_5 \, \mathrm{Cl}(K) \geq 2 \right\},$$

where Δ_K denotes the discriminant of K.

2 Known Results

Byeon [1] established the lower bound

$$N_{5,2}^{-}(X) \gg X^{1/4}.$$

Bartz, Levin, and Thamminana subsequently improved this exponent. Their result gives the current benchmark for the problem.

Theorem 1 (Bartz–Levin–Thamminana, 2025). *As $X \to \infty$,*

$$N_{5,2}^{-}(X) \gg \frac{X^{1/3}}{(\log X)^2}.$$

Their proof relates rational torsion on Jacobians of algebraic curves to ideal class groups of quadratic fields. In particular, a genus-2 curve over $\mathbf{Q}$ with sufficiently large rational 5-torsion on its Jacobian is used together with quantitative specialization arguments to construct imaginary quadratic fields satisfying

$$\mathrm{rk}_5 \, \mathrm{Cl}(K) \geq 2.$$

Thus the present state of the problem leaves a substantial gap between the known quantitative lower bound and the much stronger behavior suggested by the Cohen–Lenstra heuristics.

3 Research Problem

The main problem of this project is the following.

Problem 1. *Improve the quantitative lower bound*

$$N_{5,2}^{-}(X) \gg \frac{X^{1/3}}{(\log X)^2}.$$

The first objective is to understand the existing theorem in sufficient detail to determine which parts of the argument are responsible for the exponent $1/3$ and for the logarithmic factor $(\log X)^{-2}$. This includes reconstructing the principal theorems and lemmas used in the current proof and determining whether the quantitative losses are intrinsic or arise from particular features of the argument.

Once the present bound is understood, the project will seek any rigorous improvement. A first possible level of progress would be an improvement of the logarithmic factor, for example

$$N_{5,2}^{-}(X) \gg \frac{X^{1/3}}{(\log X)^{\alpha}}, \qquad \alpha < 2,$$

or, more strongly,

$$N_{5,2}^{-}(X) \gg X^{1/3}.$$

The principal quantitative objective is an improvement of the exponent.

Research Goal 1. *Prove that there exists an explicit $\delta > 0$ such that*

$$N_{5,2}^{-}(X) \gg X^{1/3+\delta-o(1)}.$$

The project does not assume in advance that the present method can achieve such an improvement. Determining that the exponent $1/3$ represents a genuine obstruction for a natural class of methods would itself constitute useful mathematical information.

4 Long-Term Perspective

The case of 5-rank at least two is intended to serve as the first case of a broader investigation. For an odd prime p and an integer $r \geq 1$, define

$$N_{p,r}^{-}(X) = \#\left\{K/\mathbf{Q} \text{ imaginary quadratic} : |\Delta_K| \leq X,\ \mathrm{rk}_p \, \mathrm{Cl}(K) \geq r\right\}.$$

After the case $(p, r) = (5, 2)$ is sufficiently understood, natural extensions include higher 5-rank, other odd primes such as $p = 7$, and eventually more general questions concerning the distribution of p-primary parts of quadratic class groups.

The long-term objective is therefore not restricted to a single numerical improvement. Rather, the aim is to develop a better understanding of the arithmetic mechanisms governing high p-rank class groups and to obtain increasingly strong quantitative results toward the behavior predicted by Cohen–Lenstra type heuristics.

References

[1] D. Byeon, *Quadratic fields with noncyclic* 5- *or* 7-*class groups*, The Ramanujan Journal **19** (2009), no. 1, 71–77.

[2] K. Bartz, A. Levin, and A. D. Thamminana, *Counting imaginary quadratic fields with an ideal class group of* 5-*rank at least* 2, The Ramanujan Journal **68** (2025), Article 26.